NN

Fully Connected Layer

Forward Propagation: $z = Wx + b \rightarrow z_i = \sum_j W_{ij} x_j + b_i$

Local Derivative: $g_i = \dfrac{\partial L}{\partial z_i} \rightarrow \dfrac{\partial L}{\partial W_{ij}} = g_i x_j \quad \dfrac{\partial L}{\partial x_j} = \sum_i g_i W_{ij}$

Softmax + Cross-Entropy

$$p_i = \frac{e^{z_i}}{\sum_{k=1}^{C} e^{z_k}}, \quad L = -\sum_{k=1}^{C} t_k \log p_k = -\log p_c, \quad \frac{\partial L}{\partial z_i} = p_i - t_i$$

Sigmoid + Binary Cross-Entropy

$$p = \frac{1}{1+e^{-z}}, \quad L = -[t \log p + (1-t)\log(1-p)], \quad \frac{\partial L}{\partial z} = p - t$$

Activation Functions

Activation	Function	Derivative	Advantages	Disadvantages
Sigmoid	$\dfrac{1}{1+e^{-z}}$	$f(z)(1-f(z))$	Binary Classification	Vanishing Gradients
Tanh	$\dfrac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$	$1-f(z)^2$	Zero-Centered	Vanishing Gradients
ReLU	$\max(0,z)$	$\begin{cases}1 & z>0\\0 & z<0\end{cases}$	Simple	Dying ReLU
Leaky ReLU	$\begin{cases}z & z\geq 0\\ \alpha z & z<0\end{cases}$	$\begin{cases}1 & z>0\\ \alpha & z<0\end{cases}$	Preserve Gradients	Hyperparameter α

Multi-Layer Perceptron

MLP: $z^{(l)} = W^{(l)} h^{(l-1)} + b^{(l)}, \quad h^{(l)} = \phi_l(z^{(l)})$

Activation → Introduce Nonlinearity; Linear Function → Classification Logits

Multi-Layer → Improve Representation Efficiency; Perceptron → Final Linear Classification

Regression Losses

$L_1 = \dfrac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i| \rightarrow \dfrac{\partial L_1}{\partial \hat{y}_i} = \dfrac{1}{N}\operatorname{sign}(\hat{y}_i - y_i)$ → robust to outliers

$L_2 = \dfrac{1}{N}\sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \rightarrow \dfrac{\partial L_2}{\partial \hat{y}_i} = \dfrac{2}{N}(\hat{y}_i - y_i)$ → sensitive to outliers

Batch Normalization

Mean: $\mu_j = \dfrac{1}{N}\sum_i x_{ij}$, Variance: $v_j = \dfrac{1}{N}\sum_i (x_{ij} - \mu_j)^2$

Standardization: $\bar{x}_{ij} = \dfrac{x_{ij} - \mu_j}{\sqrt{v_j + \epsilon}}$; Affine Transform: $y_{ij} = \gamma_j \bar{x}_{ij} + \beta_j$

Normalize Intermediate Features → Stabilize Gradient Scales → Larger Learning Rates

Running Statistics → Stable Batch-Independent Inference

$$g_{ij} = \frac{\partial L}{\partial y_{ij}}, \quad \frac{\partial L}{\partial \beta_j} = \sum_i g_{ij}, \quad \frac{\partial L}{\partial \gamma_j} = \sum_i g_{ij}\bar{x}_{ij}$$

$$\frac{\partial L}{\partial x_{ij}} = \frac{\gamma_j}{\sqrt{v_j+\epsilon}}\left[g_{ij} - \frac{1}{N}\sum_k g_{kj} - \frac{\bar{x}_{ij}}{N}\sum_k g_{kj}\bar{x}_{kj}\right]$$

Dropout

Ordinary Dropout: $y_i^{\text{train}} = m_i x_i, \quad y_i^{\text{test}} = q x_i$

Inverted Dropout: $y_i^{\text{train}} = m_i x_i / q, \quad y_i^{\text{test}} = x_i$

Keep Probability q; Random Mask $m_i \sim \text{Bernoulli}(q)$

Deactivate Neurons → Reduce Co-Adaptation → Mitigate Overfitting / Better Generalization

AutoEncoder

Reconstruction Loss: $h = f_\theta(x), \quad \tilde{x} = g_\phi(h), \quad J = \dfrac{1}{2N}\sum_{i=1}^{N} \left\| x^{(i)} - \tilde{x}^{(i)} \right\|_2^2$

Low-Dimensional Bottleneck → Learn Latent Representation

Backward Propagation

Gradient Descent: $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$

Newton's Method: $\theta_{t+1} = \theta_t - \left[\nabla^2 L(\theta_t)\right]^{-1} \nabla L(\theta_t)$

Practice: Multiclass NN

$x = (1,2)^\top,\ W = 0_{3\times 2},\ b = 0_3,\ c = 1,\ \eta = 0.3.$
$p = \text{softmax}(Wx + b),\ L = -\log p_1$; find $p, \nabla L, W', b'$.

Solution: $z = 0,\ p = (1,1,1)/3,\ g = p - e_1 = (-2,1,1)/3.$
$\nabla_W L = g x^\top, \quad \nabla_b L = g.$
$W' = [(.2,.4);(-.1,-.2);(-.1,-.2)], \quad b' = (.2,-.1,-.1)^\top.$

Optimization

Mini-Batch SGD

Loss Function: $J_{\mathcal{B}_t}(\theta) = \dfrac{1}{|\mathcal{B}_t|}\sum_{i\in\mathcal{B}_t} \ell_i(\theta)$

Mini-batch Gradient: $g_t = \dfrac{1}{B_t}\sum_{i\in\mathcal{B}_t} \nabla \ell_i(\theta_t), \quad B_t = |\mathcal{B}_t|$

Gradient Descent: $\theta_{t+1} = \theta_t - \eta_t g_t, \quad g_t = \nabla J_{\mathcal{B}_t}(\theta_t)$

Taylor Expansion: $J_{\mathcal{B}_t}(\theta_t - \eta_t g_t) \approx J_{\mathcal{B}_t}(\theta_t) - \eta_t g_t^\top g_t = J_{\mathcal{B}_t}(\theta_t) - \eta_t \|g_t\|_2^2$

Smaller $B_t \rightarrow$ High Gradient Variance → Noisy Updates → Improve Generalization

Larger $B_t \rightarrow$ High Memory Usage → Fewer Updates per Epoch → Better Stability

Smaller $\eta_t \rightarrow$ Slow Convergence → More Epochs

Larger $\eta_t \rightarrow$ Repeated Overshooting → Oscillation / Divergence

Momentum

Exponential Moving Average: $v_t = \mu v_{t-1} + (1-\mu) g_t \quad \theta_{t+1} = \theta_t - \eta_t v_t$

Accumulate Past Gradients → Reinforce Consistent / Reduce Oscillation → Convergence

L2 Regularization

Regularization Term: $J_{\text{reg}}(\theta) = J_{\text{data}}(\theta) + \dfrac{\lambda}{2}\|\theta\|_2^2 \quad \nabla_\theta \dfrac{\lambda}{2}\|\theta\|_2^2 = \lambda\theta$

Shrinkage Factor: $\theta_{t+1} = \theta_t - \eta(g_t + \lambda\theta_t) = (1-\eta\lambda)\theta_t - \eta g_t$

Weight Decay → Penalize Large Weights → Reduce Overfitting → Improve Generalization

Resilient Propagation

Rprop:
$$\eta_{t,i} = \begin{cases}\min(\alpha\eta_{t-1,i}, \eta_{\max}), & g_{t,i}g_{t-1,i} > 0,\ \alpha > 1\\ \max(\beta\eta_{t-1,i}, \eta_{\min}), & g_{t,i}g_{t-1,i} < 0,\ 0 < \beta < 1\\ \eta_{t-1,i}, & g_{t,i}g_{t-1,i} = 0\end{cases}$$

$$\theta_{t+1,i} = \theta_{t,i} - \eta_{t,i}\operatorname{sign}(g_{t,i})$$

Consistent Signs → Increase Step Size → Faster Convergence

Opposite Signs → Decrease Step Size → Reduce Oscillation

Adaptive Learning Rate

AdaGrad: $s_{t,i} = s_{t-1,i} + g_{t,i}^2 \quad \theta_{t+1,i} = \theta_{t,i} - \dfrac{\eta}{\sqrt{s_{t,i}+\epsilon}} g_{t,i}$

RMSProp: $s_{t,i} = \rho s_{t-1,i} + (1-\rho) g_{t,i}^2 \quad \theta_{t+1,i} = \theta_{t,i} - \dfrac{\eta}{\sqrt{s_{t,i}+\epsilon}} g_{t,i}$

Measure Gradient Magnitudes → Adapt Learning Rates → Reduce Oscillation

Adaptive Moment Estimation

First-Moment Estimate: $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \rightarrow$ Determine Update Direction

Second Raw-Moment Estimate: $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \rightarrow$ Adjust Update Step Size

Bias Correction: $\hat{m}_t = m_t/(1-\beta_1^t) \quad \hat{v}_t = v_t/(1-\beta_2^t) \rightarrow$ Correct Zero Initialization

Adam: $\theta_{t+1} = \theta_t - \eta \dfrac{\hat{m}_t}{\sqrt{\hat{v}_t}+\epsilon} \rightarrow$ Adaptive Moment Estimation

AdamW: $\theta_{t+1} = (1-\eta\lambda)\theta_t - \eta \dfrac{\hat{m}_t}{\sqrt{\hat{v}_t}+\epsilon} \rightarrow$ Decoupled Weight Decay

Gradients Problems

Vanishing / Exploding Gradients → Repeated Multiplication of Local Derivatives

Gradient Clipping: $g_{\text{clip}} = g \cdot \min\left(1, \dfrac{c}{\|g\|_2}\right) \rightarrow$ Bounded Gradient Norm

Weight Initialization

Xavier / Glorot: $W \sim \mathcal{N}\left(0, \dfrac{2}{n_{\text{in}}+n_{\text{out}}}\right) \quad W \sim U\left(-\sqrt{\dfrac{6}{n_{\text{in}}+n_{\text{out}}}}, \sqrt{\dfrac{6}{n_{\text{in}}+n_{\text{out}}}}\right) \rightarrow$ sigmoid, tanh

Kaiming / He: $W \sim \mathcal{N}\left(0, \dfrac{2}{n_{\text{in}}}\right) \quad W \sim U\left(-\sqrt{\dfrac{6}{n_{\text{in}}}}, \sqrt{\dfrac{6}{n_{\text{in}}}}\right) \rightarrow$ relu

Break Neuron Symmetry + Stabilize Signal Scale → Reduce Vanishing / Exploding

Data Augmentation

Rotation, Flipping, Cropping, Scaling, Jittering, Noise, Erasing

Learning Rate Schedules

Inverse-Time: $\eta_t = \dfrac{\eta_0}{1+dt}$, Step: $\eta_t = \eta_0 \gamma^{\lfloor t/K \rfloor}$, Exponential: $\eta_t = \eta_0 e^{-kt}$

Cosine Annealing: $\eta_t = \eta_{\min} + \dfrac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\dfrac{\pi t}{T}\right)$

Warmup: $\eta_t = \eta_{\max}\dfrac{t}{T_w}$

CNN

Advantages

Local Connectivity: Sparse Connections $\rightarrow$ Lower Computation / Local Feature Extraction

Weight Sharing: Reuse Filters $\rightarrow$ Fewer Parameters / Location-Independent Feature Detection

Translation Equivariance: $f(T_\Delta X) = T_\Delta f(X) \rightarrow$ Input Shifts, Feature Map Shifts Accordingly

Translation Invariance: $f(T_\Delta X) = f(X) \rightarrow$ Input Shifts, Classification Remains Unchanged

Forward Propagation

Feature Map: $y_{i,j,o} = b_o + \sum_{c=1}^{C_{\text{in}}} \sum_{u=0}^{k_h-1} \sum_{v=0}^{k_w-1} w_{o,c,u,v}\, x_{is_h-p_h+ud_h,\, js_w-p_w+vd_w,\, c}$

Kernel Size: $K_e = D(K-1)+1$, Max Index: $i_{max} + K - 1 \le L_{in} + 2p - 1$

Output Size: $L_{\text{out}} = \left\lfloor \dfrac{L_{\text{in}} + 2P - K_e}{S} \right\rfloor + 1$

Parameters=$(K_h K_w C_{in} + 1)\, C_{out}$, MACs=$H_{\text{out}} W_{\text{out}} C_{\text{out}} K_h K_w C_{\text{in}}$

Grouped: $C'_{\text{in}} = C_{\text{in}}/G$, Depthwise: $G = C_{\text{in}}$

Backward Propagation

Given $s = 1, p = 0, d = 1, c_{in} = 1, c_{out} = 1$

1D: $y_i = \sum_{k=0}^{K-1} w_k x_{i+k} + b \quad \delta_i = \dfrac{\partial J}{\partial y_i}$

$$\frac{\partial J}{\partial w_k} = \sum_i \delta_i x_{i+k} \quad \frac{\partial J}{\partial b} = \sum_i \delta_i \quad \frac{\partial J}{\partial x_a} = \sum_i \delta_i w_{a-i}$$

2D: $y_{i,j} = \sum_{u,v} w_{u,v} x_{i+u,j+v} + b \quad \delta_{i,j} = \dfrac{\partial J}{\partial y_{i,j}}$

$$\frac{\partial J}{\partial w_{u,v}} = \sum_{i,j} \delta_{i,j} x_{i+u,j+v} \quad \frac{\partial J}{\partial b} = \sum_{i,j} \delta_{i,j} \quad \frac{\partial J}{\partial x_{a,b}} = \sum_{i,j} \delta_{i,j} w_{a-i,b-j}$$

Receptive Field

$r_0 = 1 \quad j_0 = 1 \quad j_l = s_l j_{l-1} \quad r_l = r_{l-1} + (k_l - 1) d_l j_{l-1}$

$r_L = 1 + \sum_{l=1}^{L} (k_l - 1) d_l \prod_{m=1}^{l-1} s_m$

Pooling

Pooling	Formula	Advantages
Max	$y_{u,v,c} = \max_{(i,j)\in\Omega_{u,v}} x_{i,j,c}$	Salient Features $\rightarrow$ Improve Robustness
Average	$y_{u,v,c} = \dfrac{1}{\lvert\Omega_{u,v}\rvert} \sum_{(i,j)\in\Omega_{u,v}} x_{i,j,c}$	Smooth Features $\rightarrow$ Reduce Noise
Global Average	$y_c = \dfrac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{i,j,c}$	Aggregate Features $\rightarrow$ Reduce Parameters

BatchNorm / Dropout

$$\mu_c = \frac{1}{M} \sum_{n,h,w} x_{n,h,w,c}, \; v_c = \frac{1}{M} \sum_{n,h,w} (x_{n,h,w,c} - \mu_c)^2$$

$$y_{n,h,w,c} = \gamma_c \frac{x_{n,h,w,c} - \mu_c}{\sqrt{v_c + \epsilon}} + \beta_c, \; y_{n,h,w,c} = \frac{m_{n,c}}{1-p} x_{n,h,w,c}$$

Mask-Aware Convolutions

Partial: $y_{i,j} = \begin{cases} \frac{K}{N_{i,j}} \text{Conv}_w (x \odot m)_{i,j} + b & N_{i,j} > 0 \\ 0 & N_{i,j} = 0 \end{cases} \rightarrow$ Valid Inputs Only

Submanifold Sparse: $y_{i,j} = \begin{cases} \text{Conv}_w (x \odot m)_{i,j} + b & m_{i,j} = 1 \\ 0 & m_{i,j} = 0 \end{cases} \rightarrow$ Active Inputs Only

Modern Convolutions

Model	Key Idea	Advantages
VGG	Stacked Small Kernels	Expand Receptive Field with Less Parameters
NiN	Pointwise Convolution	Fuse Features Across Channels
GoogLeNet	Multi-Scale Convolutions	Concatenate to Capture Multi-Scale Features
ResNet	Residual Skip Connections	Enable Deeper Networks
DenseNet	Dense Connections	Enhance Feature Reuse
ResNeXt	Grouped Convolutions	Improve Representative Efficiency
MobileNet	Depthwise Separable Convolutions	Reduce Parameters / Computation
EfficientNet	Compound Scaling	Improve Accuracy–Efficiency Trade-off

Practice: CNN Backpropagation

$x = (1, 0, 2),\ w = (1, 1),\ b = 0,\ \eta = 0.1.$
$y_i = w_0 x_i + w_1 x_{i+1} + b$ for $i = 0, 1$; $L = \frac{1}{2} \sum_i y_i^2$; find gradients and updates.

Solution: $y = \delta = (1, 2),\ \delta_i = \partial L / \partial y_i.$
$\nabla_w L = (\delta_0 x_0 + \delta_1 x_1,\ \delta_0 x_1 + \delta_1 x_2) = (1, 4).$
$\nabla_x L = (\delta_0 w_0,\ \delta_0 w_1 + \delta_1 w_0,\ \delta_1 w_1) = (1, 3, 2).$
$\partial L / \partial b = \delta_0 + \delta_1 = 3, \quad w' = (.9, .6),\ b' = -.3.$

RNN

Sequence Probability Factorization

Probability Chain Rule: $P(x_{1:T}) = P(x_1) \prod_{t=2}^{T} P(x_t \mid x_{1:t-1})$

First-Order Markov Assumption: $P(x_{1:T}) = P(x_1) \prod_{t=2}^{T} P(x_t \mid x_{t-1})$

Second-Order Markov Assumption: $P(x_{1:T}) = P(x_1, x_2) \prod_{t=3}^{T} P(x_t \mid x_{t-2}, x_{t-1})$

Hidden Markov Model: $P(x_{1:T}, h_{1:T} \mid \theta) = P(h_1) \prod_{t=2}^{T} P(h_t \mid h_{t-1}) \prod_{t=1}^{T} P(x_t \mid h_t)$

Transition Distribution: $P(h_t \mid h_{1:t-1}) = P(h_t \mid h_{t-1})$

Emission Distribution: $P(x_t \mid h_{1:t}, x_{1:t-1}) = P(x_t \mid h_t)$

Vanilla Recurrent Neural Network

$a_t = W_x x_t + W_h h_{t-1} + b_h,\ h_t = \tanh(a_t),\ z_t = W_z h_t + b_z,\ p_t = \text{softmax}(z_t)$

Parameter Count: $H(I + H + 1) + O(H + 1)$

Backward Propagation Through Time

$$\ell_t = -\sum_{c=1}^{C} y_{t,c} \log p_{t,c}, \quad L = \sum_{t=1}^{T} \ell_t$$

$$\delta_t = \frac{\partial L}{\partial a_t} = \left(\frac{\partial h_t}{\partial a_t}\right)^\top \left[\left(\frac{\partial z_t}{\partial h_t}\right)^\top \frac{\partial L}{\partial z_t} + \left(\frac{\partial a_{t+1}}{\partial h_t}\right)^\top \frac{\partial L}{\partial a_{t+1}}\right]$$

$$\delta_t = \left(W_z^\top (p_t - y_t) + W_h^\top \delta_{t+1}\right) \odot (1 - h_t^2), \quad \delta_{T+1} = 0$$

$$\nabla_{W_z} L = \sum_{t=1}^{T} (p_t - y_t) h_t^\top, \quad \nabla_{W_x} L = \sum_{t=1}^{T} \delta_t x_t^\top, \quad \nabla_{W_h} L = \sum_{t=1}^{T} \delta_t h_{t-1}^\top$$

$$\nabla_{b_z} L = \sum_{t=1}^{T} (p_t - y_t), \quad \nabla_{b_h} L = \sum_{t=1}^{T} \delta_t$$

$$J_t = \frac{\partial h_t}{\partial h_{t-1}} \rightarrow \frac{\partial h_T}{\partial h_t} = J_T J_{T-1} \cdots J_{t+1}$$

Repeated Jacobian Multiplication $\rightarrow$ Vanishing / Exploding Gradients

Long Short-Term Memory

Forget Gate: $f_t = \sigma(W_f [x_t; h_{t-1}] + b_f) \rightarrow$ Old Memory to Retain

Input Gate: $i_t = \sigma(W_i [x_t; h_{t-1}] + b_i) \rightarrow$ New Info to Write

Output Gate: $o_t = \sigma(W_o [x_t; h_{t-1}] + b_o) \rightarrow$ New Memory to Expose

Candidate Cell State: $\tilde{c}_t = \tanh(W_c [x_t; h_{t-1}] + b_c) \rightarrow$ New Info

Cell State Update: $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \rightarrow$ New Memory

Hidden State: $h_t = o_t \odot \tanh(c_t) \rightarrow$ Representation

$$\left.\frac{\partial c_t}{\partial c_{t-1}}\right|_{\text{direct}} = \text{diag}(f_t) \rightarrow \left.\frac{\partial c_T}{\partial c_t}\right|_{\text{direct}} = \prod_{k=t+1}^{T} \text{diag}(f_k)$$

Separate Memory + Additive Update $\rightarrow$ Better Gradients Flow

Parameter Count: $4H(I + H + 1) + O(H + 1)$

GRU

Reset Gate: $r_t = \sigma(W_r [x_t; h_{t-1}] + b_r) \rightarrow$ Old Memory to Reuse

Update Gate: $z_t = \sigma(W_z [x_t; h_{t-1}] + b_z) \rightarrow$ Old Memory to Retain

Candidate Hidden State: $\tilde{h}_t = \tanh(W_x x_t + r_t \odot (W_h h_{t-1}) + b_h) \rightarrow$ New Info

Hidden State Update: $h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \rightarrow$ Representation

$$\left.\frac{\partial h_t}{\partial h_{t-1}}\right|_{\text{direct}} = \text{diag}(z_t) \rightarrow \left.\frac{\partial h_T}{\partial h_t}\right|_{\text{direct}} = \prod_{k=t+1}^{T} \text{diag}(z_k)$$

Unified Memory + Additive Update $\rightarrow$ Better Gradient Flow

Parameter Count: $3H(I + H + 1) + O(H + 1)$

Autoregressive Generation

$$P(x_{1:T}) = \prod_{t=1}^{T} P(x_t \mid x_{<t}) \approx \prod_{t=1}^{T} P_\theta(x_t \mid h_{t-1}) = \prod_{t=1}^{T} p_{t-1}[x_t]$$

$$L = -\log P(x_{1:T}) \approx -\sum_{t=1}^{T} \log P_\theta(x_t \mid h_{t-1}) = -\sum_{t=1}^{T} \log p_{t-1}[x_t]$$

Seq2Seq

Encoder: $v = \text{RNN}(x_1, \ldots, x_T) = h_T^{\text{enc}} = h_0^{\text{dec}}$

Decoder: $P_\theta(y_{1:T'} \mid x_{1:T}) = \prod_{t=1}^{T'} P_\theta(y_t \mid v, y_{<t}) = \prod_{t=1}^{T'} P_\theta(y_t \mid h_t^{\text{dec}}) = \prod_{t=1}^{T'} p_t[y_t]$

Practice: RNN BPTT

$a_t = w x_t + u h_{t-1} + b,\ h_t = \tanh(a_t),\ h_0 = 0.$
$(x_1, x_2) = (1, 0),\ (w, u, b) = (0, 1, 0),\ \eta = 0.1.$
$L = \frac{1}{2}(h_2 - 1)^2,\ \delta_t = \partial L / \partial a_t$; find states, gradients and updates.

Solution: $h_1 = h_2 = 0,\ \delta_2 = (h_2 - 1)(1 - h_2^2) = -1.$
$\delta_1 = u \delta_2 (1 - h_1^2) = -1.$
$\nabla_{w,u,b} L = (\delta_1 x_1 + \delta_2 x_2,\ \delta_1 h_0 + \delta_2 h_1,\ \delta_1 + \delta_2) = (-1, 0, -2).$
$(w', u', b') = (0.1, 1, 0.2).$